

The Theorem Ladder of Resolution Geometry

A Consolidated Structural Leg: What Is Forced, Declared, Imported, and Open,

from the Kernel-Pair Root to the Observable Cone

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Abstract

The corpus has grown into many separate hostile-attack papers—kernel pair, quotient, assembly, reconstruction, ignorance, observation limit, saturation, cone, scale, transport, duration. This note supplies the missing independent leg: a single dependency-ordered ladder that states, for the whole framework, exactly what forces what. It now adds one theorem-level typing correction forced by a dedicated hostile battery: the observable cone cannot jump directly to Lorentzian conformal reconstruction. A projective realization module must first separate disguised quadratic, Lorentz–Finsler, multicone, stratified, and residual characteristic branches. The remaining content is the *typing*: every rung is marked **forced** (by the exact kernel pair), **imported** (established mathematics reused), **gated** (valid only under an explicit hypothesis), **declared** (added structure, not forced), **no-go** (a forced negative), or **open** (a conjecture/gap). The spine is: an observation map F forces its kernel pair R_F , and R_F forces—and forces *only*—the exact extensional layer: observational equivalence, the quotient $Q_F \cong \text{Im } F$, fibre saturation, and universal factorization of fibre-constant constructions. Every higher structure crosses a type boundary and needs its own gate: the tangent shadow $\ker dF$ (regularity), the second-order covariance/Fisher shadow (a declared metric; Fisher provably insufficient), the statistical order (Blackwell $\neq$ kernel-pair order), assembly completeness (a sound groupoid need not generate every fibre), the observable cone (a declared fibre-compatible reachability plus a topology and a Lorentzian-realization gate, after which HKMM reconstruction gives a conformal class), scale (an integrable local clock field; a single worldline underdetermines it), and transport (the induced connection on regular strata, with the canonical connection the largest open gap). The kernel-pair gap is therefore mapped precisely: the root is universal for exact information and for nothing else, and each extension is a named module with a named hypothesis. The result is a leg one can audit rung by rung, with the frontier—connection reconstruction, Blackwell realizability, singular strata, and the physical scale anchor—marked as such.

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1 Why a leg, and how to read it

Each existing paper attacks one joint. What a foundation additionally needs is a spine: the ordered list of load-bearing steps, each tagged by *what kind* of statement it is, so a reader can see where mathematics forces the structure, where a hypothesis is silently doing the work, and where the frontier lies. The tags are:

forced	compelled by the exact kernel pair R_F alone
imported	established mathematics reused, not RG-original
gated	valid only under an explicitly stated hypothesis
declared	added structure, chosen not forced
no-go	a forced negative result (something cannot be done)
open	conjecture or gap; not established

The organizing fact, established by the kernel-pair-root attack, is that only the first stratum below is *forced*; every later rung crosses a type boundary.

2 The ladder

Rung	Statement	Type	Depends on
<i>Stratum 0 — the exact root (the only forced layer)</i>			
R0	Observation map $F : X \rightarrow Y$ declared	input	—
R1	Kernel pair $R_F = X \times_Y X$ is an equivalence relation (exact ignorance)	forced	R0
R2	Quotient $Q_F = X/R_F \cong \text{Im } F$; every fibre-constant map factors uniquely through Q_F	forced	R1
R3	Saturation: observable invariants are exactly R_F -invariant functions on Q_F	forced	R2
<i>Stratum 1 — assembly (bottom-up = top-down, conditionally)</i>			
R4	Kernel groupoid $K_{\text{obs}} = X \times_{Q_F} X$; R_F is a groupoid	forced	R1
R5	Declared automorphism groupoid A ; sound $A \hookrightarrow K_{\text{obs}}$ gives surjection $X/A \rightarrow Q_F$	gated (soundness)	R4
R6	$X/A \cong Q_F$ iff A is gauge-complete (generates every fibre)	gated (completeness)	R5
<i>Stratum 2 — second-order shadow (metric detection)</i>			
R7	Regular reconstruction: (R, N, Σ) is a canonical gauge-equivariant function of (J, Σ, g)	imported	R2
R8	Fisher = $J^\top \Sigma^{-1} J$ = inverse Schur = second-order tangent shadow	imported	R7
R9	Fisher insufficiency: blind to the null block at $C_{RN} = 0$; does not reconstruct the covariance	no-go	R8

Rung	Statement	Type	Depends on
R10	Tangent caveat: $\ker dF = \text{tangent of the fibre only on regular strata; higher contact is jets of } F, \text{ not covariance}$	gated (regularity)	R2,R8
<i>Stratum 3 — ignorance and the observation limit</i>			
R11	On regular models $\text{rad } g^F = \ker dF$ (Fisher detects, does not define, ignorance)	gated (regularity)	R8,R10
R12	Persistent null $\mathcal{I}_x = \ker dF_\infty$ over a compatible sensor family; attained finitely (finite dim)	gated (compatibility)	R11
R13	Three layers: sensor = wall $\mathcal{I}_x$; precision (evaluation) and representation (downstream) = removable	forced (typing)	R12
R14	Kernel-pair order $\neq$ Blackwell order; equal identifiability kernel pairs need not be Blackwell-equivalent	no-go	R1
<i>Stratum 4 — declared structure on the quotient</i>			
R15	Fibre-compatible reachability $\preceq$ descends to a preorder on Q_F ; antisymmetrize to a causal order	declared	R2
R16	Future sets J_{obs}^+ ; a cone boundary requires a supplied compatible topology	gated (topology)	R15
R17	Projective-cone invariance: classify the zero set/direction boundary, not the degree or chosen defining function x	no-go / typing	R16
R18	Quadratic-cone gate: equality with the null cone of a nondegenerate Lorentzian quadratic form gives a unique positive conformal ray $[g_{\text{obs}}]$	imported (gated)	R17
R19	Lorentz–Finsler gate: a single smooth, time-oriented, strongly convex homogeneous cone with nondegenerate Lorentzian fibre Hessian admits a direction-dependent realization	gated	R17
R20	Multicone gate: persistent multiple projective characteristic sheets require a multicone or multi-Finsler realization; one conformal metric is insufficient	gated	R17
R21	Stratified-realization gate: rank, isotropy, signature, topology, or sheet-count changes require regular strata joined by slices/tangent cones; failures of homogeneity or convexity remain general characteristic/admissibility geometry	gated / open globally	R17–R20
R22	Scale: $\mathcal{T} = -\log C$ additive on the zero-memory stratum; a single worldline underdetermines the conformal factor	gated	R17
R23	Metric representative exists iff local scale data glue as an exact cocycle (zero scale-holonomy)	gated (integrability)	R22

Rung	Statement	Type	Depends on
R24	Cross-stratum (astronomy): remote null test after a correspondence map and nuisance model; ratio-only identifiability	gated	R17,R23
<i>Stratum 5 — transport, and the main open gap</i>			
R25	On a regular metric stratum with projector P : $\nabla^W = P \nabla^{\text{LC}}$; second form $\Pi = (I - P) \nabla^{\text{LC}}$	imported (gated)	R7,R17
R26	Entropy no-go: $H(X), H(Q), H(X \mid Q)$ are fibre-automorphism invariant; entropy cannot select a connection	no-go	R2
R27	Canonical connection from (J, Σ, K) (memory kernel required): $(J, \Sigma, K) \rightarrow [\nabla]$	open	R25,R26
<i>Stratum 6 — duration algebra (self-contained)</i>			
R28	$\mathcal{T} = -\log C$; memory = BCH defect $\frac{1}{2}[\mathcal{T}, \mathcal{T}] + \dots$; additive iff commuting	imported	R22
R29	Scalar clocks = trace character $\tau = a \operatorname{tr} \mathcal{T}$, unique up to unit (Cauchy)	imported	R28

3 The kernel-pair gap, stated exactly

The single most important structural fact is where the root stops.

Gap 1 (What R_F forces, and what it does not). The exact kernel pair forces exactly R1–R4: observational equivalence, the quotient, fibre saturation, universal factorization, and the kernel groupoid. It forces *nothing else*. Each of the following is a separate type and requires its own hypothesis:

- (i) the *tangent shadow* $\ker dF$ equals the fibre tangent only on regular strata (R10); at singular points it does not.
- (ii) *higher contact* is governed by jets of F , not by any covariance; formal flatness need not imply exact local constancy (R10).
- (iii) *Fisher/covariance* is a declared distinguishability structure, not kernel-pair data, and is insufficient to reconstruct the covariance (R8–R9).
- (iv) the *statistical order* (Blackwell/deficiency) is strictly finer than sameness of identifiability kernel pair (R14).
- (v) *assembly* closes only under gauge completeness; a sound groupoid need not generate every fibre (R6).
- (vi) *precision* and *representation* act on evaluation and on downstream constructions, not on R_F (R13).

These six crossings are the kernel-pair gap. Naming them is the leg’s central service: it forbids smuggling any higher module into the root.

Gap 2 (The main open rung). R27—a canonical connection from (J, Σ, K) —is the largest remaining source of non-uniqueness in the characterized object. The entropy no-go (R26) shows a natural candidate selector fails; the memory kernel (R28) is the extra datum the reconstruction

needs; the theorem is not yet in hand. Everything downstream of a connection (curvature, holonomy, stratified transport) inherits this gap.

Gap 3 (The physical anchor is outside the ladder). Absolute scale (R22–R23) and the identification $v_\partial = c$ are declared/empirical, not forced by any rung. This is why the cone yields $[g]$ but not g : the ladder is conformal-complete and scale-open by construction. The anchor is a physical input, correctly placed outside the forced spine.

4 The minimal “must” path to the observable cone

Extracting only the load-bearing rungs, the shortest sound route from an experiment to an observable characteristic geometry is

$$\underbrace{F}_{\text{R0}} \rightarrow \underbrace{R_F \rightarrow Q_F}_{\text{R1–R2, forced}} \rightarrow \underbrace{\preceq_{\text{obs}}}_{\text{R15, declared}} \rightarrow \underbrace{\text{topology and cone boundary}}_{\text{R16, gate}} \rightarrow \underbrace{\text{projective realization test}}_{\text{R17}}.$$

The realization test branches rather than continuing automatically:

$$\begin{array}{lll} \text{quadratic regular cone} & \xrightarrow{\text{R18}} & [g_{\text{obs}}], \\ \text{single smooth strongly convex cone} & \xrightarrow{\text{R19}} & \text{Lorentz–Finsler target}, \\ \text{persistent multiple sheets} & \xrightarrow{\text{R20}} & \text{multicone target}, \\ \text{type-changing discriminant} & \xrightarrow{\text{R21}} & \text{stratified tangent-cone/slice target}. \end{array}$$

If homogeneity, regularity, or convexity fails, R21 retains only general characteristic or admissibility geometry. Groupoid assembly (R4–R6) remains an alternative certificate for R2. The second-order stratum (R7–R11) is required only for metric/ignorance geometry. Scale (R22–R24) and transport (R25–R27) are further modules. The quadratic path is therefore not universal; it is one gated branch of a larger realization rung.

5 Final cone-realization attack and reproducibility

The deterministic artifact `rg_cone_realization_final_attack.py`, seed 20260925, registered and passed all 13/13 hostile checks. The machine-readable record is `rg_cone_realization_final_attack.json`. The attacks were:

1. a degree-six presentation $x = Q^3$ with exactly the same projective null cone as a Lorentzian quadratic form, rejecting polynomial degree as a realization invariant;
2. a smooth homogeneous single-sheet witness that rejects every fitted quadratic cone on held-out directions while retaining a positive spatial fibre Hessian and Lorentzian fundamental-tensor signature, forcing the Lorentz–Finsler branch;
3. a product of two separated Lorentzian factors, producing two persistent characteristic sheets that cannot be represented by one nondegenerate Lorentzian quadratic form, forcing the multicone branch;
4. a parameterized sheet collision whose sheet count changes $2 \rightarrow 1 \rightarrow 2$ and whose characteristic gradient vanishes at the collision wall, forcing a stratified transition locus;
5. nonhomogeneous and homogeneous-but-nonconvex residual witnesses, proving that the named branches are gated and are not an exhaustive theorem without further hypotheses.

The analytic conclusion is stronger than a numerical fit: the explicit witnesses are counterexamples to the previous universal jump from R16 to Lorentzian conformal reconstruction. Hence R17–R21 are logically necessary typing rungs. The battery does *not* prove that Finsler, multicone, and stratified geometry exhaust every possible characteristic object; the residual branch is retained explicitly.

6 Non-claims

Non-claim 1 (No exhaustive classification theorem). The new realization module proves that an intermediate rung is necessary and supplies explicit witnesses for its principal branches. It does not claim that Lorentz–Finsler, multicone, and stratified cones exhaust all homogeneous characteristic geometry. The residual branch in R21 is intentional.

Non-claim 2 (No new primitive). The root is a kernel pair; the imported rungs are Schur/Fisher, HKMM reconstruction, BCH, and Cauchy’s equation. No rung introduces a primitive unknown to standard mathematics.

Non-claim 3 (Conformal, not metric; observational, not dynamical). The forced and gated rungs reach a conformal class and an observational transport at most. Absolute scale, the light-speed identification, and Einstein dynamics are outside the ladder.

Non-claim 4 (Gaps are gaps). R14 (Blackwell), R21 (global stratified compatibility), R27 (canonical connection), and the physical anchor are open or external and are not asserted as closed.

7 Conclusion

The corpus has an independent leg: a spine of thirty rungs in seven strata, each typed, with the exact kernel pair forcing only the first four and every later structure carrying a named hypothesis. The observable cone is reached by a short forced-plus-gated path; the metric, the transport, and the physics are further modules, each gated or open. The two structural facts worth carrying are that the kernel pair is universal for exact information *and nothing else*—the six type crossings are the gap—and that the single largest remaining non-uniqueness is the canonical connection R27. A framework that can state its own dependency graph, its imports, and its open rungs this precisely is one whose claims can be audited; that auditability, not any new theorem, is what this leg supplies.

References

- [1] M. Rada, *The Kernel-Pair Root and Its Limits*, 2026.
- [2] M. Rada, *Resolution Geometry as an Observable Orbit–Cone Structure*, 2026.
- [3] M. Rada, *Resolution Geometry: Reconstruction and Identifiability*, 2026.
- [4] M. Rada, *The Law and Boundary of Observational Ignorance*, 2026.
- [5] M. Rada, *Observation-Constrained Causal Order and Conformal Cone Geometry*, 2026.
- [6] S. W. Hawking, A. R. King, P. J. McCarthy, *J. Math. Phys.* 17 (1976), 174.
- [7] D. B. Malament, *J. Math. Phys.* 18 (1977), 1399.
- [8] M. J. Pflaum, H. Posthuma, X. Tang, *J. reine angew. Math.* 694 (2014), 49.

- [9] L. Le Cam, *Comparison of Experiments*, IMS, 1996.
- [10] F. Zhang (ed.), *The Schur Complement and Its Applications*, Springer, 2005.